

EARTHSCAPE CLIMATE AGENCY

TABLE OF CONTENTS

EARTHSCAPE CLIMATE AGENCY

Credential.....

Admin Guide

1. Login Page

2. Dashboard.....

3. Dataset Management

4. Job Management.....

5. Climate Explorer:.....

6. Alerts Management

7. User Management

8. Support Requests

9. Help

10. Logout

2

3

4

4

5

6

7

8

9

10

11

12

13

Earthscape Climate Agency

Supervisor	Sir Wajahat	
Batch	2309f	
Project Title	Earthscape Climate Agency	
Serial no.	Enrollment no.	Student name
1.	1500503	M. Masoom
2.	1476532	Hamna Sarfaraz

Credential

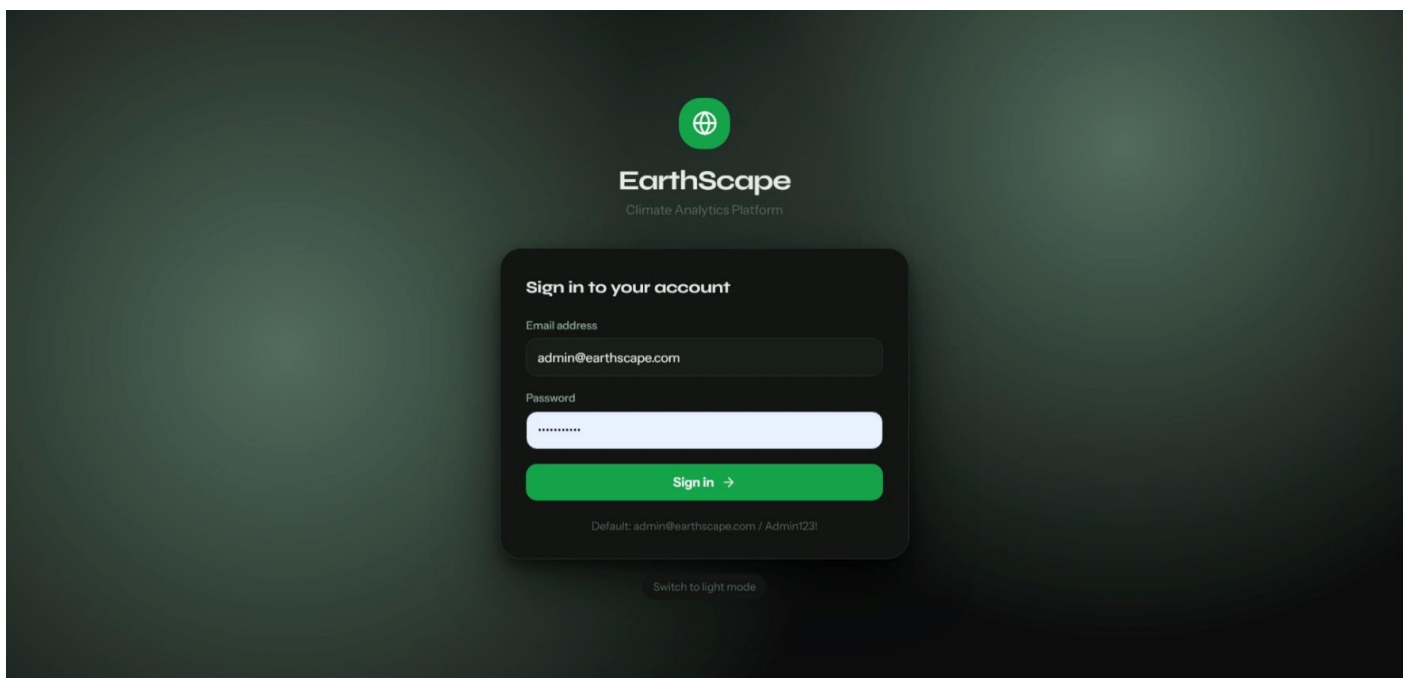
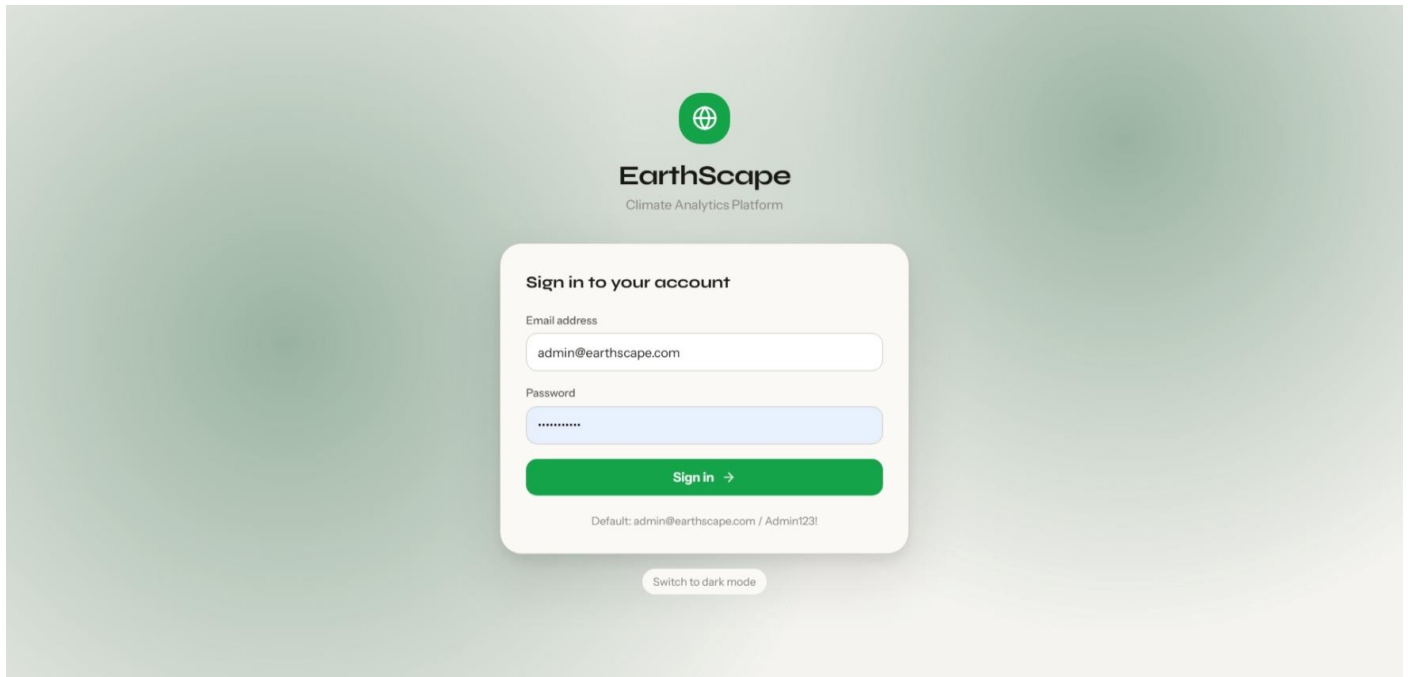
Admin Credential:

- Username: admin@earthscape.com
 - Password: Admin123!
-

Admin Guide

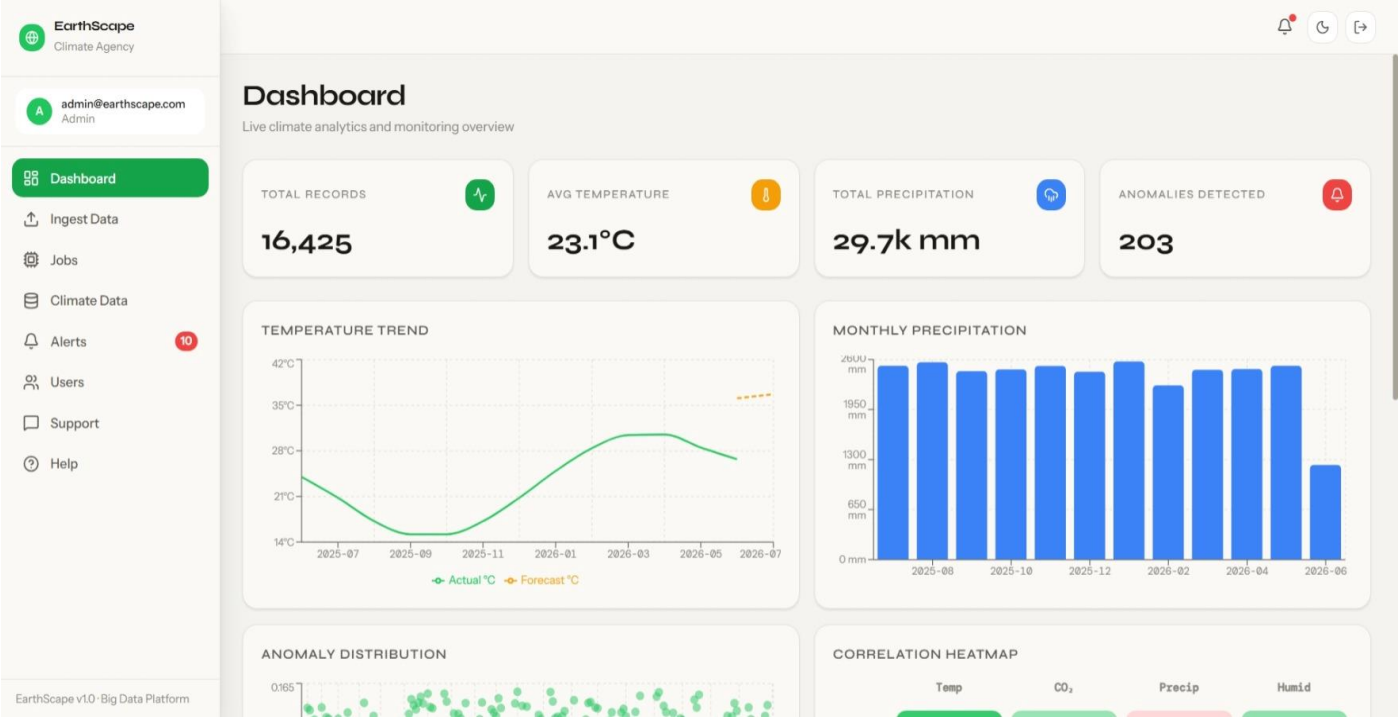
1. Login Page

- Secure login for authorized administrators only.
- Default credentials provided for first-time access.
- Option to switch between light and dark mode.



2. Dashboard

- Displays **key metrics**: total records, average temperature, total precipitation, anomalies detected.
- Includes **visual charts**: temperature trends, precipitation bars, anomaly distribution, correlation heatmaps.



3. Dataset Management

- Admins can **add, update, or delete categories** (Weather Station, Satellite, Sensor).
- Upload datasets (CSV, JSON, GeoJSON up to 50 MB).
- Ingestion logs show file name, format, record count, status, and timestamp.

The screenshot shows the EarthScape v1.0 Big Data Platform interface. The left sidebar contains navigation links: Dashboard, Ingest Data (highlighted), Jobs, Climate Data, Alerts (with a red badge showing 10), Users, Support, and Help. The main content area is titled 'Data Ingestion' with the subtitle 'Upload climate datasets to HDFS'. It features an 'UPLOAD DATASET' section with a dashed box for file upload and a 'Weather Station' dropdown menu. Below this is an 'INGESTION LOG' section with a 'Refresh' button and a table showing the upload of 'seed_data.csv' in CSV format with 16,425 records, successfully ingested on Jun 16, 2026, at 08:22 PM.

EarthScape v1.0: Big Data Platform

This screenshot shows the same EarthScape v1.0 Big Data Platform interface as the previous one, but with a dark theme. The layout and content are identical, including the sidebar navigation, the 'Data Ingestion' header, the upload area, and the ingestion log table showing the successful upload of 'seed_data.csv'.

EarthScape v1.0: Big Data Platform

4. Job Management

- Trigger **MapReduce jobs** (Temperature Aggregation, Precipitation Totals).
- Train **ML models** (Anomaly Detection, Trend Prediction, Correlation Analysis).
- Job history table shows name, type, status, duration, and start time.

The screenshot shows the 'Jobs' management interface in the EarthScape v1.0 - Big Data Platform. The left sidebar contains navigation links: Dashboard, Ingest Data, Jobs (highlighted), Climate Data, Alerts (14), Users, Support, and Help. The main content area is titled 'Jobs' and includes a subtitle 'Trigger MapReduce processing and ML model training'.

There are two main configuration panels:

- MAPREDUCE JOB:** Includes fields for Job Name (test 2), Job Type (Temperature Aggregation (CSV)), and Input File (HDFS) - CSV only (seed_data.csv - 16425 records - Jun 8, 2026, 06:36 AM). A 'Submit MapReduce Job' button is at the bottom.
- ML TRAINING:** Includes a Model Type dropdown (Anomaly Detection (Isolation Forest)) and a 'Start Training' button. A note states: 'Training reads from MongoDB climate_records and writes predictions back. Results visible on dashboard.'

Below these panels is a 'JOB HISTORY' table with columns: NAME, TYPE, STATUS, DURATION, and STARTED. The table lists 10 jobs, all with a status of 'completed'.

NAME	TYPE	STATUS	DURATION	STARTED
ml_anomaly_detection	ml_train	completed	0s	Jun 25, 2026, 08:52 PM
ml_correlation	ml_train	completed	0s	Jun 21, 2026, 07:57 AM
ml_trend_prediction	ml_train	completed	1s	Jun 21, 2026, 07:57 AM
ml_anomaly_detection	ml_train	completed	1s	Jun 21, 2026, 07:56 AM
test 1	mapreduce	completed	2s	Jun 21, 2026, 07:56 AM
ml_correlation	ml_train	completed	0s	Jun 8, 2026, 06:36 AM
ml_trend_prediction	ml_train	completed	0s	Jun 8, 2026, 06:36 AM
ml_anomaly_detection	ml_train	completed	0s	Jun 8, 2026, 06:36 AM
Precipitation Totals - Seed Run	mapreduce	completed	9s	Jun 8, 2026, 05:06 AM
Temperature Aggregation - Seed Run	mapreduce	completed	12s	Jun 8, 2026, 04:36 AM

This screenshot shows the same 'Jobs' management interface as the first image, but in a dark theme. The layout and content are identical, including the sidebar, configuration panels for MapReduce and ML Training, and the Job History table.

The 'JOB HISTORY' table in the dark theme is as follows:

NAME	TYPE	STATUS	DURATION	STARTED
ml_correlation	ml_train	completed	0s	Jun 21, 2026, 07:57 AM
ml_trend_prediction	ml_train	completed	1s	Jun 21, 2026, 07:57 AM
ml_anomaly_detection	ml_train	completed	1s	Jun 21, 2026, 07:56 AM
test 1	mapreduce	completed	2s	Jun 21, 2026, 07:56 AM
ml_correlation	ml_train	completed	0s	Jun 8, 2026, 06:36 AM
ml_trend_prediction	ml_train	completed	0s	Jun 8, 2026, 06:36 AM
ml_anomaly_detection	ml_train	completed	0s	Jun 8, 2026, 06:36 AM
Precipitation Totals - Seed Run	mapreduce	completed	9s	Jun 8, 2026, 05:06 AM
Temperature Aggregation - Seed Run	mapreduce	completed	12s	Jun 8, 2026, 04:36 AM

5. Climate Explorer:

- View climate datasets by region (Karachi, Lahore, Islamabad) and source (Weather Station, Satellite, Sensor).
- Records table shows temperature, precipitation, humidity, CO₂ levels, anomaly status, and timestamp.
- Filter datasets using Region, Source, Date Range, and Anomaly status.
- Export filtered datasets as CSV files.
- Records flagged as anomalies are highlighted for immediate review.

The screenshot shows the EarthScape Climate Explorer interface in a light theme. The left sidebar contains navigation links: Dashboard, Ingest Data, Jobs, Climate Data (highlighted), Alerts (10), Users, Support, and Help. The main area is titled 'Climate Explorer' with the subtitle 'Browse and export climate records'. It features a filter bar with 'Region' (Karachi), 'Source' (All), 'From' and 'To' date pickers (dd/mm/yyyy), and 'Anomalies' (All), followed by an 'Apply' button. Below the filters is a table of records with 16,425 total entries. The table columns are REGION, SOURCE, TEMP °C, PRECIP MM, HUMIDITY %, CO₂ PPM, ANOMALY, and TIMESTAMP. The data shows records for Karachi, Lahore, and Islamabad from various sources (weather_station, satellite, sensor) with timestamps from June 15, 2026. One record for Islamabad from a weather_station is marked as an anomaly (Yes).

REGION	SOURCE	TEMP °C	PRECIP MM	HUMIDITY %	CO ₂ PPM	ANOMALY	TIMESTAMP
Karachi	weather_station	29.3	2.23	71.3	416.9	No	Jun 15, 2026, 04:00 PM
Karachi	satellite	29.6	0.00	70.9	413.6	No	Jun 15, 2026, 04:00 PM
Karachi	sensor	31.1	2.27	69.5	415.6	No	Jun 15, 2026, 04:00 PM
Lahore	weather_station	26.3	2.39	58.8	418.8	No	Jun 15, 2026, 04:00 PM
Lahore	satellite	28.0	2.19	62.1	417.3	No	Jun 15, 2026, 04:00 PM
Lahore	sensor	26.0	2.50	61.8	417.5	No	Jun 15, 2026, 04:00 PM
Islamabad	weather_station	21.9	4.04	53.7	410.6	Yes	Jun 15, 2026, 04:00 PM
Islamabad	satellite	23.3	3.84	57.1	411.5	No	Jun 15, 2026, 04:00 PM

This screenshot shows the same EarthScape Climate Explorer interface but in a dark theme. The layout and data are identical to the light theme version, including the sidebar, filter bar, and the table of climate records. The 'Climate Data' link in the sidebar and the 'Apply' button are highlighted in green. The table shows the same 16,425 records with columns for region, source, temperature, precipitation, humidity, CO₂ levels, anomaly status, and timestamp. The Islamabad weather_station record is again marked as an anomaly.

REGION	SOURCE	TEMP °C	PRECIP MM	HUMIDITY %	CO ₂ PPM	ANOMALY	TIMESTAMP
Karachi	weather_station	29.3	2.23	71.3	416.9	No	Jun 15, 2026, 04:00 PM
Karachi	satellite	29.6	0.00	70.9	413.6	No	Jun 15, 2026, 04:00 PM
Karachi	sensor	31.1	2.27	69.5	415.6	No	Jun 15, 2026, 04:00 PM
Lahore	weather_station	26.3	2.39	58.8	418.8	No	Jun 15, 2026, 04:00 PM
Lahore	satellite	28.0	2.19	62.1	417.3	No	Jun 15, 2026, 04:00 PM
Lahore	sensor	26.0	2.50	61.8	417.5	No	Jun 15, 2026, 04:00 PM
Islamabad	weather_station	21.9	4.04	53.7	410.6	Yes	Jun 15, 2026, 04:00 PM
Islamabad	satellite	23.3	3.84	57.1	411.5	No	Jun 15, 2026, 04:00 PM

6. Alerts Management

- Create new rules (metric, operator, threshold, severity).
- Manage alert rules: enable, disable, delete.
- View triggered events with severity, value, timestamp, and status.

The screenshot shows the EarthScape Alerts Management interface in a light theme. The left sidebar contains navigation links: Dashboard, Ingest Data, Jobs, Climate Data, Alerts (highlighted with a red badge showing '10'), Users, Support, and Help. The main content area is titled 'Alerts' with a subtitle 'Manage alert rules and review triggered events'. A '+ New Rule' button is in the top right. Below the title is a section for 'ALERT RULES' containing a table with columns: NAME, CONDITION, SEVERITY, STATUS, and ACTIONS. The table lists five rules: High Temperature Warning, Low Temperature Alert, Heavy Precipitation, Elevated CO₂, and Low Humidity. Each rule has a 'Disable' button and a delete icon. Below this is a section for 'ALERT EVENTS' with a 'Refresh' button and a table with columns: SEVERITY, VALUE, TRIGGERED, STATUS, and ACTION. The table shows one event with severity 'high', value '45.5', triggered at 'Jun 16, 2026, 08:22 PM', and status 'Acknowledged'. The footer indicates 'EarthScape v1.0 - Big Data Platform'.

NAME	CONDITION	SEVERITY	STATUS	ACTIONS
High Temperature Warning	temperature_c > 40	high	Active	Disable
Low Temperature Alert	temperature_c < 5	medium	Active	Disable
Heavy Precipitation	precipitation_mm > 50	medium	Active	Disable
Elevated CO ₂	co2_ppm > 430	low	Active	Disable
Low Humidity	humidity_pct < 20	low	Active	Disable

SEVERITY	VALUE	TRIGGERED	STATUS	ACTION
high	45.5	Jun 16, 2026, 08:22 PM	Acknowledged	

This screenshot shows the same EarthScape Alerts Management interface but in a dark theme. The layout and content are identical to the light theme version, including the sidebar, navigation links, and the 'Alerts' management section with tables for alert rules and events. The footer also reads 'EarthScape v1.0 - Big Data Platform'.

NAME	CONDITION	SEVERITY	STATUS	ACTIONS
High Temperature Warning	temperature_c > 40	high	Active	Disable
Low Temperature Alert	temperature_c < 5	medium	Active	Disable
Heavy Precipitation	precipitation_mm > 50	medium	Active	Disable
Elevated CO ₂	co2_ppm > 430	low	Active	Disable
Low Humidity	humidity_pct < 20	low	Active	Disable

SEVERITY	VALUE	TRIGGERED	STATUS	ACTION
high	45.5	Jun 16, 2026, 08:22 PM	Acknowledged	

7. User Management

- Manage platform access and roles (Admin, Analyst, Viewer).
- Add new users or deactivate existing ones.
- User table shows email, role, status, created date, and actions.

The screenshot shows the 'Users' management interface in a light-themed application. The left sidebar contains navigation links: Dashboard, Ingest Data, Jobs, Climate Data, Alerts (with a red badge showing 10), Users (highlighted in green), Support, and Help. The main content area is titled 'Users' with the subtitle 'Manage platform access and roles'. A green '+ Add User' button is in the top right. Below the title is a table labeled 'ALL USERS' with 3 total entries. The table has columns for EMAIL, ROLE, STATUS, CREATED, and ACTIONS. The data rows are as follows:

EMAIL	ROLE	STATUS	CREATED	ACTIONS
analyst@earthscope.com	analyst	Active	Jun 16, 2026	Deactivate
viewer@earthscope.com	viewer	Active	Jun 16, 2026	Deactivate
admin@earthscope.com	admin	Active	Jun 16, 2026	Deactivate

The footer of the page reads 'EarthScape v1.0 - Big Data Platform'.

This screenshot shows the same 'Users' management interface as the first image, but in a dark-themed version. The layout and content are identical, including the sidebar with the 'Users' link highlighted in green, the '+ Add User' button, and the table listing three users (analyst, viewer, and admin) with their roles, active status, and creation dates. The footer also reads 'EarthScape v1.0 - Big Data Platform'.

8. Support Requests

- Submit and track tickets.
- Ticket table shows subject, status (open, in-progress, resolved), submitted date, and actions.
- Admins can respond, update status, and provide feedback.

The screenshot shows the EarthScape Support interface in a light theme. On the left is a sidebar with navigation links: Dashboard, Ingest Data, Jobs, Climate Data, Alerts (with a red badge showing 10), Users, Support (highlighted in green), and Help. The main content area is titled 'Support' with the subtitle 'Submit and track support requests'. A '+ New Ticket' button is in the top right. Below the title is a 'TICKETS' section containing a table with three rows of tickets. Each row has columns for SUBJECT, STATUS, SUBMITTED, and ACTIONS. The first ticket is 'Request access to Jobs page' with status 'open'. The second is 'Export CSV cap exceeded' with status 'in-progress'. The third is 'Dashboard charts not loading' with status 'resolved'. Each row has a 'Respond' button in the ACTIONS column.

SUBJECT	STATUS	SUBMITTED	ACTIONS
Request access to Jobs page	open	Jun 16, 2026, 02:22 PM	Respond
Export CSV cap exceeded Response: Apply region and date filters to stay under the 10k cap...	in-progress	Jun 15, 2026, 08:22 PM	Respond
Dashboard charts not loading Response: Ensure at least 90 days of climate records are ingested befo...	resolved	Jun 13, 2026, 08:22 PM	Respond

This screenshot shows the same EarthScape Support interface as the first image, but in a dark theme. The layout and content are identical, including the sidebar with the 'Support' link highlighted, the 'Support' header, and the table of three tickets with their respective statuses and actions.

SUBJECT	STATUS	SUBMITTED	ACTIONS
Request access to Jobs page	open	Jun 16, 2026, 02:22 PM	Respond
Export CSV cap exceeded Response: Apply region and date filters to stay under the 10k cap...	in-progress	Jun 15, 2026, 08:22 PM	Respond
Dashboard charts not loading Response: Ensure at least 90 days of climate records are ingested befo...	resolved	Jun 13, 2026, 08:22 PM	Respond

9. Help

- This section assists administrators in managing EarthScape's backend features.

EarthScape
Climate Agency

admin@earthscape.com
Admin

Dashboard
Ingest Data
Jobs
Climate Data
Alerts
Users
Support
Help

EarthScape v10: Big Data Platform

Help

User guide for the EarthScape platform

Getting Started

- Sign in at the login page using your email and password. Your session is stored in a secure httpOnly cookie and expires after 60 minutes of inactivity. Click **Logout** in the top-right corner to end your session immediately.
- First run only: `views@earthscape.com / Views1231`

Data Ingestion

- Navigate to **Ingest Data** in the sidebar (Analyst or Admin only).
- Select the **Source Type** — Weather Station, Satellite, or Sensor.
- Drag and drop a file into the upload zone, or click to browse.
- Supported: **CSV, JSON, GeoJSON** — max 50 MB per file.
- Files with the same content (SHA-256 hash) are rejected automatically.
- Each upload creates a log entry: `pending` → `success` / `failed`. Expand any log row to see the HDFS path and record count.

Jobs

- Go to **Jobs** in the sidebar.
- Select a job type first, then choose an input file — only files matching that job type's required format (CSV or JSON) are shown.
- Temperature Aggregation** (CSV) — min / max / mean per region
- Precipitation Totals** (CSV) — monthly totals per region
- Anomaly Scores** (JSON) — Z-score based per-record anomaly rating
- Click **Submit MapReduce Job**. Status updates every 5 seconds.
- Anomaly Detection** — Isolation Forest, last 90 days
- Trend Prediction** — Linear Regression, last 365 days, 30-day forecast
- Correlation Analysis** — Pearson matrix, last 365 days
- Click **Start Training**. Results appear on the Dashboard charts automatically.

Alerts

- Go to **Alerts** and click **New Rule** (Admin only).
- Choose a metric, comparison operator, threshold value, and severity.
- The rule evaluates against every incoming sensor record in real time.
- Rules can be enabled, disabled, or deleted from the rules table.
- Any authenticated user can acknowledge an alert event. Click **Ack** on any pending event to remove it from the notifications panel.

Support

- Go to **Support** and click **New Ticket**.
- Subject, a detailed description, and optionally a screenshot URL.
- Track status (Open → In Progress → Resolved) in the ticket list.
- Admins can respond to any ticket via the **Respond** button, update the status, and add a written response visible to the submitting user.

Tips

- The Dashboard auto-refreshes in real time — no manual reload needed.
- Use the Climate Explorer filter bar to narrow large datasets before exporting.
- CSV exports are capped at 10,000 records per request. Apply filters to stay under the cap.
- The live feed panel in the dashboard shows the 10 most recent sensor readings.

EarthScape
Climate Agency

admin@earthscape.com
Admin

Dashboard
Ingest Data
Jobs
Climate Data
Alerts
Users
Support
Help

EarthScape v10: Big Data Platform

Help

User guide for the EarthScape platform

Getting Started

- Sign in at the login page using your email and password. Your session is stored in a secure httpOnly cookie and expires after 60 minutes of inactivity. Click **Logout** in the top-right corner to end your session immediately.
- First run only: `views@earthscape.com / Views1231`

Data Ingestion

- Navigate to **Ingest Data** in the sidebar (Analyst or Admin only).
- Select the **Source Type** — Weather Station, Satellite, or Sensor.
- Drag and drop a file into the upload zone, or click to browse.
- Supported: **CSV, JSON, GeoJSON** — max 50 MB per file.
- Files with the same content (SHA-256 hash) are rejected automatically.
- Each upload creates a log entry: `pending` → `success` / `failed`. Expand any log row to see the HDFS path and record count.

Jobs

- Go to **Jobs** in the sidebar.
- Select a job type first, then choose an input file — only files matching that job type's required format (CSV or JSON) are shown.
- Temperature Aggregation** (CSV) — min / max / mean per region
- Precipitation Totals** (CSV) — monthly totals per region
- Anomaly Scores** (JSON) — Z-score based per-record anomaly rating
- Click **Submit MapReduce Job**. Status updates every 5 seconds.
- Anomaly Detection** — Isolation Forest, last 90 days
- Trend Prediction** — Linear Regression, last 365 days, 30-day forecast
- Correlation Analysis** — Pearson matrix, last 365 days
- Click **Start Training**. Results appear on the Dashboard charts automatically.

Alerts

- Go to **Alerts** and click **New Rule** (Admin only).
- Choose a metric, comparison operator, threshold value, and severity.
- The rule evaluates against every incoming sensor record in real time.
- Rules can be enabled, disabled, or deleted from the rules table.
- Any authenticated user can acknowledge an alert event. Click **Ack** on any pending event to remove it from the notifications panel.

Support

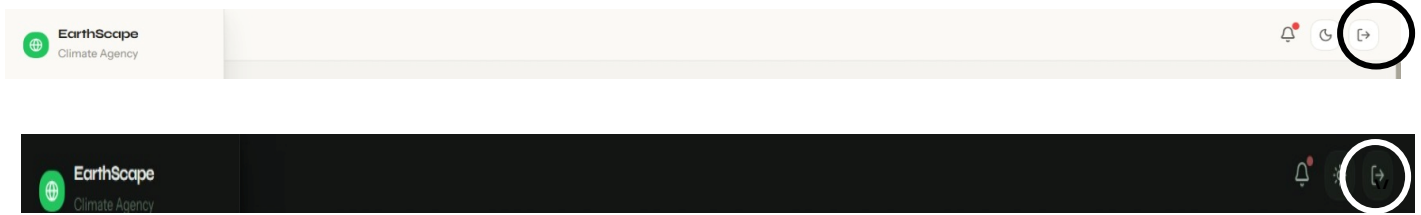
- Go to **Support** and click **New Ticket**.
- Subject, a detailed description, and optionally a screenshot URL.
- Track status (Open → In Progress → Resolved) in the ticket list.
- Admins can respond to any ticket via the **Respond** button, update the status, and add a written response visible to the submitting user.

Tips

- The Dashboard auto-refreshes in real time — no manual reload needed.
- Use the Climate Explorer filter bar to narrow large datasets before exporting.
- CSV exports are capped at 10,000 records per request. Apply filters to stay under the cap.
- The live feed panel in the dashboard shows the 10 most recent sensor readings.

10. Logout

- Securely exit the admin panel via the logout button.
- Ends session immediately.



End of Admin Guide.
THANK YOU!
